T-99000: C4MBI67 Primitive-Prefix Reconstruction
Repository deterministic core-font rendering

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Equations may be displayed in TeX syntax in this portability rendering.
Scientific status: the exact finite theorem through 10^9 is proved; the all-scale C4MBI67 sign and the Riemann Hypothesis remain unproved.

The exact theorem ledger

Statement	Status
Source dictionary and Mobius inverse	proved exactly
Primitive-prefix/window identity	proved exactly
Four-band C4MBI67 identity	proved exactly
Logarithmic prime ownership	proved exactly
Weighted threshold-complex representation	proved exactly
Windowed Bellman recurrence	proved exactly
Mellin transform and off-line-zero noncancellation	proved exactly
Specialized Landau theorem	proved below
Theta-functional-equation symmetry	proved below
$C(t) \ge 0$ for $1 \le t < 10^9 + 1$	exact rational certificate
$\mathcal{A}_X \ge 0$ for all sufficiently large X	open / RH-bearing
Riemann Hypothesis	unproved

Arithmetic reconstruction from the source dictionary

Mobius inversion

The Mobius function is defined by $\mu(1) = 1$, by $\mu(n) = 0$ if a prime square divides n, and by $\mu(p_1 \dots p_k) = (-1)^k$ for distinct primes.

Lemma 1 (Dirichlet inverse of the constant-one function). For every positive integer n,
$\sum_{d \mid n}\mu(d) = 1_{n = 1}$.

Proof. If $n = 1$ the sum is 1. If $n > 1$ has distinct prime divisors $p_1, \dots, p_r$, only squarefree divisors contribute and the sum is
$\sum_{S \subset \{1, \dots, r\}}(-1)^{|S|} = (1 - 1)^r = 0.$
QED

Define the positive sharp source dictionary
$q(1) = 0, \quad q(2) = 15, \quad q(4) = 3, \quad q(n) = 6 \quad (n \notin \{1, 2, 4\}).$
Let $a = q * \mu$ under Dirichlet convolution.

Lemma 2 (Explicit native coefficients). For every $n \ge 1$,
$a(n) = 6 1_{n = 1} - 6\mu(n) + 9 1_{2 \mid n}\mu(n/2) - 3 1_{4 \mid n}\mu(n/4).$

Proof. Write
$q = 6 1 - 6\delta_1 + 9\delta_2 - 3\delta_4,$
where $1(n) = 1$ and $\delta_j(n) = 1_{n = j}$. Convolution with μ, using $1 * \mu = \delta_1$, gives
$q * \mu = 6\delta_1 - 6\mu + 9(\delta_2 * \mu) - 3(\delta_4 * \mu),$
and $(\delta_j * \mu)(n) = 1_{j \mid n}\mu(n/j)$. QED

For later checking, if $n = 2^e m$ with m odd and squarefree, then except for $a(1) = 0$,
$a(m) = -6\mu(m), \quad a(2m) = 15\mu(m), \quad a(4m) = -12\mu(m), \quad a(8m) = 3\mu(m),$
and $a(2^e m) = 0$ for $e \ge 4$ or nonsquarefree m.

The annular scalar and its primitive prefix

For $y > 0$ write $\log y_+ = \max(\log y, 0)$ and define
$$R_X=\sum_{n\ge1}\frac{a(n)}{\sqrt n}\log(X/n)_+,
\qquad
\mathcal A_X=R_X-R_{X/4}.$$
All sums are finite. Also put
$H_X(n) = \min\{\log 4, \log(X/n)_+\}.$
Then directly

$$\label{eq:annular-discrete}
 \mathcal A_X=\sum_{n\ge1}\frac{a(n)}{\sqrt n}H_X(n).$$

Definition 3 (Primitive prefix). For t > 0 set
$$C(t)=\sum_{n\le t}\frac{a(n)}{\sqrt n},
 \qquad C(t)=0\quad(0<t<1).$$

Theorem 4 (Exact primitive-prefix window). For every real X > 0,
$$\label{eq:Rwindow}
 R_X=\int_1^X C(t)\frac{dt}{t},
 \qquad
 \boxed{\mathcal A_X=\int_{X/4}^{X}C(t)\frac{dt}{t}}.$$

Proof. For each n,
$$\log(X/n)_+=\int_1^X\mathbf 1_{n\le t}\frac{dt}{t}.$$
Only n <= X occur, so finite Fubini gives the first identity. Subtracting the same identity at X/4 gives the second, with the convention C(t) = 0 for t < 1. QED

Let
$$B_{1/2}(t)=\sum_{n\le t}\frac{\mu(n)}{\sqrt n},
 \qquad B_{1/2}(t)=0\quad(t<1).$$

Theorem 5 (Three-band primitive prefix). For every real t > 0,
$$\label{eq:Ddef}
 C(t)=6-\mathcal D(t),
 \qquad
 \mathcal D(t)=6B_{1/2}(t)-\frac9{\sqrt2}B_{1/2}(t/2)
            +\frac32B_{1/2}(t/4).$$

Proof. Sum [eq:aexplicit] over n <= t with weight n^(-1/2). The three shifted sums are
$$\sum_{2m\le t}\frac{\mu(m)}{\sqrt{2m}}
 =\frac1{\sqrt2}B_{1/2}(t/2),\qquad
 \sum_{4m\le t}\frac{\mu(m)}{\sqrt{4m}}
 =\frac12B_{1/2}(t/4).$$
The 6delta_1 term is 6 for t >= 1 and both sides vanish appropriately below 1. QED

Corollary 6 (A stronger pointwise producer). If C(t) >= 0 for every sufficiently large t, and the finitely many earlier windows are checked separately, then Acal_(X) >= 0 for every sufficiently large X. In particular, global C(t) >= 0 implies global Acal_(X) >= 0.

Proof. Use the positive measure dt/t in [eq:Rwindow]. QED

C4MBI67 and exact four-band form

Substitution of [eq:aexplicit] into [eq:annular-discrete] gives
Acal_(X) = 6H_(X)(1) + Cfrak_(X),
where
$$\mathfrak C_X=\sum_{n\ge1}\frac{\mu(n)}{\sqrt n}\kappa_X(n),
\qquad
 \kappa_X(n)=-6H_X(n)+\frac9{\sqrt2}H_{X/2}(n)
            -\frac32H_{X/4}(n).$$
Thus C4MBI67 is the inequality
Cfrak_(X) >= -6H_(X)(1),
which is exactly Acal_(X) >= 0.

Theorem 7 (Exact four-band boundary). For every X > 0, with B_(1/2)(t) = 0 below 1,
$$\begin{aligned}
\mathfrak C_X={}&-\frac32\int_{X/16}^{X/8}B_{1/2}(t)\frac{dt}{t}
+\frac{9\sqrt2-3}{2}\int_{X/8}^{X/4}B_{1/2}(t)\frac{dt}{t}\notag\\
&+\frac{9\sqrt2-12}{2}\int_{X/4}^{X/2}B_{1/2}(t)\frac{dt}{t}
-6\int_{X/2}^{X}B_{1/2}(t)\frac{dt}{t}.\label{eq:fourband}
\end{aligned}$$

Proof. From Theorems 4 and 5,
$$\mathfrak C_X=-\int_{X/4}^{X}\mathcal D(t)\frac{dt}{t}.$$
In the second and third terms of [eq:Ddef], substitute u = t/2 and u = t/4. Partition the resulting

intervals at X/8, X/4, X/2 and collect coefficients. They are respectively
$$-\frac32,\quad -\frac32+\frac9{\sqrt2},\quad
 -6+\frac9{\sqrt2},\quad -6,$$
which are exactly those in [eq:fourband]. QED

Exact logarithmic prime ownership

Lemma 8 (Squarefree logarithmic ownership). For every squarefree integer n > 1,
$$\label{eq:owneratom}
 \mu(n)= -\sum_{p\mid n}\mu(n/p)\frac{\log p}{\log n},$$
where the sum is over the distinct prime divisors of n.

Proof. Since n is squarefree, mu(n/p) = -mu(n) for every p | n, and
$$-\sum_{p\mid n}\mu(n/p)\frac{\log p}{\log n}
 =\mu(n)\frac{\sum_{p\mid n}\log p}{\log n}=\mu(n).$$
QED

Theorem 9 (Prime-Mobius owner decomposition). For every finitely supported kernel kappa,
$$\label{eq:owner}
\sum_{n>1}\frac{\mu(n)}{\sqrt n}\kappa(n)
=-\sum_p\frac{\log p}{\sqrt p}
 \sum_{\substack{m\ge1\\p\nmid m}}
 \frac{\mu(m)}{\sqrt m}\frac{\kappa(pm)}{\log(pm)}.$$

Proof. Insert [eq:owneratom] in the left side, interchange two finite sums, and write n = pm. The condition p m is precisely squarefree ownership. The weights log p/log n are nonnegative and sum to one for each source atom n. QED

A weighted threshold complex

The Dirichlet series of a is especially rigid. For Rez > 1,
$$\begin{aligned}
 Q(z)&=\sum_{n\ge1}\frac{q(n)}{n^z}
 =6\zeta(z)-6+9\,2^{-z}-3\,4^{-z},\\
 A(z)&=\sum_{n\ge1}\frac{a(n)}{n^z}
 =\frac{Q(z)}{\zeta(z)}
 =6-\frac{3(1-2^{-z})(2-2^{-z})}{\zeta(z)}.
\end{aligned}$$
Let d be defined by 6delta_1 - a = d; then $\mathcal D(t)=\sum_{n\le t}d(n)/\sqrt n$. Writing z = s + 1/2 gives
$$\label{eq:Euler-generalized}
 \frac16\sum_{n\ge1}\frac{d(n)}{n^{s+1/2}}
 =(1-2^{-s-1/2})^2(1-2^{-s-3/2})
  \prod_{p\,\mathrm{odd}}(1-p^{-s-1/2}).$$

Definition 10 (Generalized-prime labels). There are two labels 2_1, 2_2 of cost 2 and activity 2^(-1/2), one label 2_3 of cost 2 and activity 2^(-3/2), and one label of cost p and activity p^(-1/2) for every odd prime p. For a finite labelled subset S, let P(S) and r(S) be the products of its costs and activities.

Theorem 11 (Weighted Euler characteristic). For every t >= 1,
$$\begin{aligned}
 \frac{\mathcal D(t)}6&=\sum_{S:\,P(S)\le t}(-1)^{|S|}r(S),\label{eq:Z}\\
 \frac{C(t)}6&=\sum_{\varnothing\ne S:\,P(S)\le t}
            (-1)^{|S|+1}r(S).\label{eq:F}
\end{aligned}$$
If each available label is independently retained with probability equal to its activity and K_(t)[R] is the induced threshold complex, then
$$\label{eq:expectedchi}
 \boxed{\frac{C(t)}6=\mathbb E\,\chi(K_t[R]).}$$

Proof. For a fixed t, only finitely many labels can occur. Expanding the finite product in [eq:Euler-generalized] gives [eq:Z] coefficient by coefficient; [eq:F] follows from C = 6 - D. Finally,
Echi(K_(t)[R]) = SUM_(empty != S: P(S) <= t)(-1)^(|S| + 1)Pr (S subset R),
and independence gives Pr (S subset R) = r(S). QED

This reformulation is not a positivity proof: a threshold complex can have negative Euler characteristic. It does, however, identify the exact orientation information absent from an unsigned energy estimate.

Exact windowed Bellman recurrence

For a finite label set V, set

$$Z_V(t)=\sum_{S\subset V,\ P(S)\le t}(-1)^{|S|}r(S),\qquad Z_V(t)=0\ (t<1),$$

$$F_V(t)=\mathbf 1_{t\ge1}-Z_V(t),
 \qquad G_V(X)=\int_{X/4}^{X}F_V(t)\frac{dt}{t}.$$

Theorem 12 (One-label Bellman update). If a new label has cost p and activity r, then

$$Z_{V\cup\{p\}}(t)=Z_V(t)-rZ_V(t/p),$$

and

$$\label{eq:Grec}
 \boxed{G_{V\cup\{p\}}(X)
 =G_V(X)+r\bigl(H_X(p)-G_V(X/p)\bigr)}.$$

For the complete generalized-prime set, $G(X)=\mathcal A_X/6$.

Proof. Split subsets in [eq:Z] according to whether the new label is absent or present; the latter are $\{p\}\cup S$ with $P(S)\le t/p$, proving [eq:Zrec]. Since $Z_V(y)=1_{y\ge1}-F_V(y)$,

$$F_{V\cup\{p\}}(t)=F_V(t)+r(1_{t/p\ge1}-F_V(t/p)).$$

Integrate over $[X/4,X]$ and substitute $u=t/p$ in the last term. The indicator integral is exactly $H_X(p)$. The final assertion follows from Theorems 4 and 11. QED

Equation [eq:Grec] is the sharp scalar obstruction. If a child satisfies $G_V(X/p)>H_X(p)$, adjoining the label contributes negatively. Any proof that replaces this correlated child by an unsigned reservoir loses the sign needed at the root.

The Mellin transform

Lemma 13 (Elementary logarithmic Mellin integral). For $n>0$ and $\operatorname{Re}s>0$,

$$\int_n^\infty \log(X/n)X^{-s-1}\,dX=\frac{n^{-s}}{s^2}.$$

Proof. Put $X=ne^{u}$. The integral becomes $n^{-s}\int_0^{\infty}ue^{-su}\,du$. Integration by parts gives $1/s^2$. QED

Because $a(n)=O(1)$, one has $\mathcal A_X=O(\sqrt X)$; hence its Mellin transform converges absolutely for $\operatorname{Re}s>1/2$.

Theorem 14 (Exact zero-safe transform). For $\operatorname{Re}s>1/2$ and $z=s+1/2$,

$$\label{eq:Mellin}
 \boxed{
 \int_1^\infty\mathcal A_X X^{-s-1}\,dX
 =\frac{1-4^{-s}}{s^2}
  \left[6-\frac{3(1-2^{-z})(2-2^{-z})}{\zeta(z)}\right].}$$

Proof. Finite truncation followed by absolute convergence allows termwise use of Lemma 13:

$$\int_1^\infty R_X X^{-s-1}dX=\frac{A(z)}{s^2}.$$

The substitution $X=4Y$ shows that the transform of $R_{X/4}$ is $4^{-s}A(z)/s^2$. Insert the explicit $A(z)$ above. QED

If ρ is a zeta zero with $\operatorname{Re}\rho>1/2$, then at $s_0=\rho-1/2$ none of the factors

$$1-4^{-s_0},\qquad 1-2^{-\rho},\qquad 2-2^{-\rho}$$

vanishes: their possible zeros have real parts 0, 0, -1, respectively. Thus [eq:Mellin] has a nonremovable pole at s_0 of the same order as the zero of zeta.

A self-contained Landau theorem

Theorem 15 (Landau singularity for an eventually nonnegative Laplace density). Let f be locally integrable and of exponential order on $[0,\infty)$. Suppose $f(t)\ge0$ for all sufficiently large t. Let

$$F(s)=\int_0^{\infty}f(t)e^{-st}\,dt$$

have finite abscissa of convergence σ_c. Then F cannot be analytic in a neighborhood of the real point $s=\sigma_c$.

Proof. Subtract the compactly supported initial part of f; its Laplace transform is entire, so we may assume $f \geq 0$ everywhere. Suppose F were analytic in a disc about σ_c. Choose real $\sigma_1 > \sigma_c$ inside that disc and $r > 0$ so that $\sigma_1 - r < \sigma_c$ and the Taylor series of F at σ_1 converges at $\sigma_1 - r$. Differentiation under the absolutely convergent integral gives
$(-1)^k F^{(k)}(\sigma_1) = \int_0^{\infty} t^k e^{-\sigma_1 t} f(t)\, dt \geq 0$.
Tonellis theorem, applicable because every summand is nonnegative, yields

$$\begin{aligned}
F(\sigma_1-r)
&=\sum_{k\ge0}\frac{(-r)^k}{k!}F^{(k)}(\sigma_1)\\
&=\int_0^\infty e^{-\sigma_1t}f(t)
\sum_{k\ge0}\frac{(rt)^k}{k!}\,dt\\
&=\int_0^\infty e^{-(\sigma_1-r)t}f(t)\,dt<\infty.
\end{aligned}$$

This contradicts the definition of σ_c. QED

The Mellin version follows by $X = e^t$:
$\int_1^{\infty} \mathcal{A}_X X^{-s-1} dX = \int_0^{\infty} \mathcal{A}_{e^t} e^{-st} dt$.

The zeta facts needed by the consumer

We include the standard analytic continuation and symmetry to make the consumer independent of an unlisted import.

Lemma 16 (Gaussian Fourier transform). For $t > 0$, the Fourier transform of $x \; e^{-\pi t x^2}$ is $t^{-1/2} e^{-\pi^2/t}$.

Proof. Scaling reduces to $t = 1$. If $I() = \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x} dx$, differentiation and integration by parts give $I() = -2\pi I()$. Since $I(0) = 1$ (the squared Gaussian integral in polar coordinates), $I() = e^{-\pi^2}$. QED

Lemma 17 (Jacobi theta identity). For $t > 0$,
$\theta(t) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t} = t^{-1/2}\theta(1/t)$.

Proof. Periodize the Gaussian: $P(x) = \sum_{n \in \mathbb{Z}} e^{-\pi t(x + n)^2}$. Uniform convergence with all derivatives permits its Fourier series. The mth Fourier coefficient is the Gaussian Fourier transform at m, namely $t^{-1/2} e^{-\pi m^2/t}$. Evaluating the Fourier series at $x = 0$ gives the identity. QED

For $\mathrm{Re} z > 1$, termwise Mellin integration gives

$$\label{eq:LambdaIntegral}
\Lambda(z):=\pi^{-z/2}\Gamma(z/2)\zeta(z)
=\frac12\int_0^\infty(\theta(t)-1)t^{z/2-1}dt.$$

Split at 1, apply Lemma 17 on $(0, 1)$, and put $u = 1/t$. One obtains

$$\label{eq:LambdaContinue}
\Lambda(z)=\frac1{z-1}-\frac1z+\frac12\int_1^\infty(\theta(t)-1)
\left(t^{z/2-1}+t^{(1-z)/2-1}\right)dt.$$

The integral is entire because $\theta(t) - 1$ decays exponentially. Formula [eq:LambdaContinue] gives the meromorphic continuation and is unchanged by $z \; 1 - z$. Hence
$(z) = (1 - z)$.
Since Gamma has no zeros, every zero in $0 < \mathrm{Re} z < 1$ is reflected to a zero at $1 - z$.

For real $z > 1$, the absolutely convergent Euler product $\zeta(z) = \prod_p (1 - p^{-z})^{-1}$ is positive and nonzero. For $0 < z < 1$, the alternating series
$\eta(z) = \sum_{n \geq 1} (-1)^{n-1} n^{-z} = \sum_{k \geq 1} ((2k - 1)^{-z} - (2k)^{-z})$
converges and is strictly positive. The identity $\eta(z) = (1 - 2^{1-z})\zeta(z)$, first obtained for $z > 1$ by separating even terms and then continued through [eq:LambdaContinue], shows that $\zeta(z) < 0$ on $(0, 1)$. Thus zeta has no positive real zero.

The complete conditional implication

Theorem 18 (C4MBI67 implies RH). If $\mathcal{A}_X \geq 0$ for all sufficiently large real X, then every nontrivial zero of zeta has real part 1/2.

Proof. Let $F(s)$ denote the Mellin transform in [eq:Mellin]. Since $\mathcal A_X=O(\sqrt X)$, its abscissa of convergence is finite and at most 1/2. After removing a compact interval, the density is nonnegative, so Theorem 15 says that the real abscissa sigma_(c) must be a singularity.

Assume a zero rho has beta = Rerho > 1/2. The noncancellation following Theorem 14 gives a pole of F at s_0 = rho - 1/2; hence the integral cannot converge in a neighborhood of s_0, and sigma_(c) >= beta - 1/2 > 0. Landau therefore forces a singularity at the positive real point sigma_(c).

But the right side of [eq:Mellin] is analytic at every real s > 0. For s > 1/2 this follows from the Euler product; for 0 < s < 1/2 it follows from the absence of real zeros on 1/2 < z < 1; and at s = 1/2 the pole of zeta makes 1/zeta(z) vanish rather than blow up. This is a contradiction. Therefore no zero has real part greater than 1/2.

If a nontrivial zero had real part below 1/2, the functional equation [eq:functional] would produce one above 1/2. Hence all nontrivial zeros lie on the critical line. QED

Corollary 19. Global primitive-prefix positivity C(t) >= 0 implies RH.

Proof. Combine Corollary 6 and Theorem 18. QED

In the project Bellman coordinates, if $U_{\rm full}=U_Z-I_Z-T_Z$ and BLPTE67 is T_(Z) <= U_(Z) - I_(Z), then
$$\mathrm{BLPTE67}(X)\Longleftrightarrow U_{\rm full}(X)\ge0.$$
At the root $U_{\rm full}(X)=\mathcal A_X/\sqrt X$, so this is exactly C4MBI67. This last equivalence is algebra, not an additional analytic lemma.

Exact rational certificate through one billion

The enclosure

Set S = 2^4^0. For each n <= 10^9, define
$$q_n=\left\lfloor\frac{S}{\sqrt n}\right\rfloor.$$
The checker verifies, using unsigned 128-bit integer multiplication,
q_(n)^2n <= S^2 < (q_(n) + 1)^2n.
Therefore
$$\frac{q_n}{S}\le\frac1{\sqrt n}<\frac{q_n+1}{S}.$$
For a(n) > 0 the lower contribution is a(n)q_(n)/S; for a(n) < 0 it is a(n)(q_(n) + 1)/S. The opposite choices give an upper enclosure. All cumulative bounds are exact signed 128-bit integers.

The Mobius sieve

The source uses a standard linear least-prime-factor recurrence. If l(n) is the least prime factor, then
$$\mu(np)=\begin{cases}0,&p=\ell(n),\\-\mu(n),&p<\ell(n),\end{cases}$$
and each composite is generated exactly once by its least prime factor. The array stores l(n) in 16 bits. A prime larger than 65534 is represented by the sentinel 65535. This is exact on the certified range: every composite n <= 10^9 has a prime factor at most $\sqrt n\le31623<65535$, while if n itself is a larger prime, every p with np <= 10^9 is smaller than the sentinel. Thus the linear-sieve comparison is unchanged.

Overflow is also excluded a priori. In [eq:qcert], the largest temporary is below 2^1^1^0, within unsigned 128-bit range. A crude bound on a cumulative signed numerator is 15 * 10^9 * 2^4^0 < 2^7^5, within signed 128-bit range.

Theorem 20 (Billion-endpoint finite theorem). For every integer 2 <= N <= 10^9,
$$\label{eq:minbound}
 C(N)\ge\frac{1226685126915}{2^{40}}
 =1.1156636236728445510379970073699951171875\ldots>0.$$
The smallest lower enclosure occurs at N = 48433, where
$$\label{eq:mininterval}
 \frac{1226685126915}{2^{40}}
 \le C(48433)\le
 \frac{1226685458193}{2^{40}}.$$
Consequently C(t) >= 0 for all real 1 <= t < 10^9 + 1, and Acal_(X) >= 0 for all real 1 <= X < 10^9 + 1.

Proof. The checker performs the exact recurrence in Lemma 2, the sieve just proved correct, and the interval update following [eq:qcert]. It compares the lower cumulative numerator after every integer endpoint. The smallest exact integer is 1226685126915, at 48433. Because $C(t)$ is constant on each half-open interval $[N, N + 1)$, the integer result is all-real. The final assertion follows by integrating the nonnegative prefix in Theorem 4. QED

The retained run reports 759, 908, 859 nonzero coefficient updates and uses no floating value in the verdict. A floating center is printed only as a human readability diagnostic.

Hostile audit: what still has to be proved

Why a PSD completion is not enough

A symmetric completion
$$\begin{pmatrix}\Gamma&A\\A&\Gamma\end{pmatrix}\succeq0$$
proves $|A| \le \Gamma$, not $A \ge 0$. The matrix with $\Gamma = 1$ and $A = -1/2$ is positive definite and has negative off-diagonal. Hence an exact Tao/Schur magnitude port still needs an oriented cross-scale comparison.

Why the finite theorem is not an asymptotic tail

The barrier in [eq:Ddef] is the fixed number 6. Classical zero-free regions give cancellation estimates for $B_{(1/2)}(t)$ that are subexponential relative to $\sqrt t$, but those upper bounds still grow without bound and cannot be compared to 6 for all large t. Thus no standard PNT error term extends Theorem 20 to infinity.

The exact remaining producer

Any one of the following would close the route:

1. Acal_(X) >= 0 for every sufficiently large real X (exact C4MBI67);

2. the stronger $C(t) \ge 0$ for every sufficiently large t together with a finite initial check;

3. an oriented Bellman invariant for [eq:Grec] that controls child overovershoot without replacing it by absolute mass;

4. an equivalent one-sided signed Type-II estimate retaining the owner weights in Theorem 9.

None is proved here. Every prior lemma needed after such a producer has been reconstructed above.

Conclusion

The T-97701 route now has a single, sharply visible mathematical boundary. The arithmetic dictionary, primitive prefix, four-band transform, prime ownership, Bellman recurrence, Mellin numerator, Landau theorem, and zeta symmetry have all been proved in one document. The exact finite theorem is extended to one billion endpoints with a rational certificate. The remaining assertion is not a bookkeeping gap: it is the all-scale oriented sign Acal_(X) >= 0. Until that producer is proved, RH remains unproved.

Replay commands

From the packet root:

```
cd experiments/X-99000-c4mbi67-primitive-prefix-billion
python3 verify.py
python3 -m unittest discover -s tests -v
# Optional full regeneration (requires about 3.2 GB RAM):
./build_full.sh 1000000000
cd ../..
sha256sum -c SHA256SUMS
```